

Stavan Rupareliya

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Professional Summary

Experienced Engineer with 4.6+ years of hands-on expertise in C/C++, Python, and hardware-software integration. Proficient in debugging and performance profiling. Skilled in hardware acceleration, model compression, and system-level performance tuning. Passionate about transforming cutting-edge research into scalable, production-grade solutions for edge intelligence.

Tech-Stacks

Technology: Embedded Systems, AI/ML, Design Verification, Artificial Intelligence, Edge Computing

Development Languages: Verilog, C/C++, Python

Operating Systems : Embedded Linux, Linux, Windows

AI/ML Frameworks: PyTorch, TensorFlow, TFLite, Keras, Caffe, NumPy, Pandas, Matplotlib

Debugging Tools: GDB, Valgrind

Version Control & CI/CD : Git, GitHub, Docker

Soft Skills: Multi-tasking, Adaptability, Problem-solving, Mentoring

Professional Experience

Tata Elxsi Limited

Mar. 2025 - Present

Senior Software Engineer

Bangalore, Karnataka.

- Developed AI/ML-based text and image summarization pipelines for multimedia applications on embedded platforms. Using deep learning models for efficient content understanding and compression.
- Deployed and optimized AI inference pipelines on edge devices, focusing on performance profiling, debugging, and resource optimization within embedded Linux environments.

elnfochips (An Arrow Company)

July 2021 - Mar. 2025

Software Engineer

Ahmadabad, Gujarat

- Designed and implemented a real-time video analytics system by integrating AI/ML models with embedded hardware platforms. Led hardware-software co-design efforts to ensure seamless integration between embedded systems and AI pipelines.
- Applied model optimization techniques such as quantization and pruning to enhance inference efficiency while maintaining high accuracy. Developed a modular camera simulator with a customizable image processing pipeline, enabling rapid prototyping and validation of client Proofs of Concept (PoCs).

elnfochips (An Arrow Company)

Jan. 2021 - July 2021

Project Trainee Engineer

Ahmadabad, Gujarat

- Completed training in C++, Python, embedded systems, and Linux internals with hands-on system program ming and memory management projects. Contributed to client deliverables and assisted in building future project PoCs.

Technical Publications

- The visual computer, An International Journal of Computer Graphics, Springer publication, 2020 “**Real-time detection and identification of plant leaf diseases using convolutional neural networks on an embedded platform**”, Springer Publication. [Paper link](#)
- 2nd International Conference on Communication, Computing and Industry 4.0 (C2I4) , 2021. “**Real-time Face Mask Detection System on Edge using Deep Learning and Hardware Accelerators**”, IEEE Publication. [Paper link](#)
- Proceedings of the International e-Conference on Intelligent Systems and Signal Processing, 2021. “**Real-Time Tomato Detection, Classification, and Counting System Using Deep Learning and Embedded Systems**”, Springer Publication [Paper link](#)
- Advances in VLSI, Communication, and Signal Processing, 2022 “**Face Mask Detection and Counting Using Deep Learning and Embedded Systems**”, Springer Publication. [Paper link](#)

Technical Projects

Synchronous FIFO Design

Tech-Stack: Verilog, Linux

Description: Implemented a Synchronous FIFO (First-In-First-Out) memory buffer in Verilog, commonly used for data buffering between modules operating under the same clock domain. The project focused on robust design and verification of control logic, status flags, and boundary conditions.

Smart Parking Management System

Tech-Stack: Verilog, Git, Linux

Description: Design and simulate a Smart Parking System that monitors entry and exit of vehicles, displays available parking slots, and controls the gate based on slot availability — using FSM logic in Verilog.

Video Analytics System Using Edge Computing

Tech-Stack: C/C++, Python, Linux, NVIDIA DeepStream SDK, Edge-AI Device

Description: Developed an edge-based video analytics system using C++ and Python to process real-time video streams for use cases such as Person Counting, Face Recognition, Fire & Smoke Detection, Object Detection, Color Clothing Identification, Vehicle Detection, Crowd Counting, and Fall Detection enabling low-latency inference on edge devices and reducing cloud dependency.

Multi-Camera Surveillance System for Autonomous Vehicles

Tech-Stack: Embedded Linux, Python, Computer Vision, GStreamer

Description: Designed and deployed a multi-camera video processing system with advanced use cases including Driver Monitoring, Lane Detection, and Pedestrian Detection to improve autonomous vehicle safety. Achieved real-time processing of up to 4 camera streams using GStreamer and OpenCV, optimizing inference on the i.MX95 NPU with latency under 200 ms per frame.

Guest Lectures & Industry Engagements

- Delivered an Industry Expert Lecture on **Embedded AI & Computer Vision Applications** at Nirma University, Institute of Technology, covering real-time AI deployment, embedded optimization, and edge intelligence use cases.
- Industry Expert Lecture on **Edge Computing and Computer Vision** Invited as an industry expert to deliver a technical session for M.Tech students specializing in VLSI and Embedded Systems.
- **Expert Lecturer**, Nirma University FDP — Delivered a session on Edge Computing and Computer Vision representing eInfochips (An Arrow Company).

Awards & Recognitions

Core Value Award - Customer First

Jul. 2024

- Recognized for exceptional dedication to prioritizing customer needs, resulting in increased customer satisfaction and loyalty. Demonstrated an outstanding ability to address customer concerns and ensure a positive customer experience.

Pat On The Back Award

Mar. 2022

- Honored for exemplary performance and significant contributions to image processing project. Demonstrated exceptional leadership, teamwork, and problem-solving skills, consistently exceeding expectations.

Educational Qualifications

Master of Technology, Embedded Systems Engineering

Jun. 2018 – May 2020

Nirma University, Institute of Technology, India.

Bachelor of Engineering, Instrumentation and Control Engineering

Jun. 2012 – May 2016

Gujarat Technological University (GTU), India.